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7 Proteins can control cell processes and determine cell structure.

(a) Molecules of pre-mRNA are produced by transcription of active genes.

These molecules undergo post-transcriptional changes to form active mRNA.

Describe how post-transcriptional changes of pre-mRNA can produce active mRNA.

(3)

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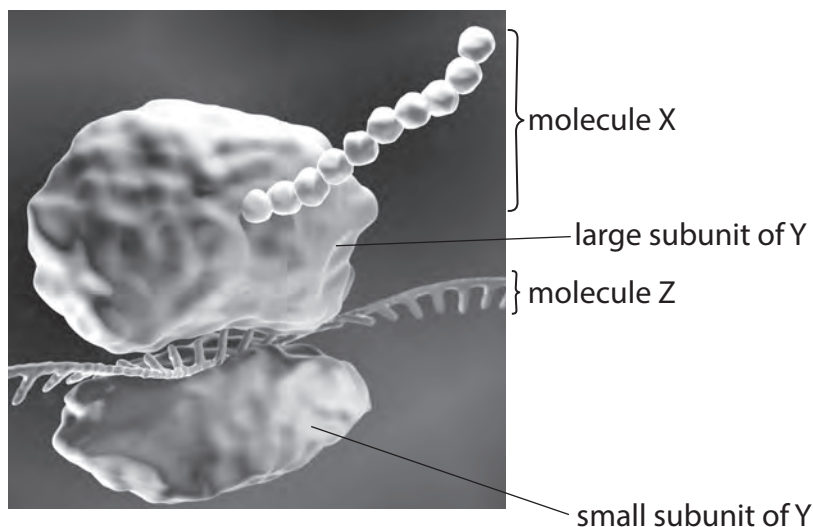
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(b) The diagram shows a process taking place on structure Y in a eukaryotic cell.



(Source: © Artur Plawgo / Science Photo Library)

- (i) Structure Y can be found in the cytoplasm or on the surface of an organelle in a eukaryotic cell.

Which organelle would have structure Y on its surface?

(1)

- ☐ **A** Golgi apparatus
- ☐ **B** mitochondria
- ☐ **C** rough endoplasmic reticulum
- ☐ **D** smooth endoplasmic reticulum

- (ii) Give the function of structure Y.

(1)

- (iii) Which row gives the names of the molecules labelled X and Z?

(1)

	Molecule X	Molecule Z
☐ A	polypeptide	DNA strand
☐ B	polypeptide	RNA strand
☐ C	polysaccharide	DNA strand
☐ D	polysaccharide	RNA strand



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(c) DNA methylation can alter the activation of a gene.

- (i) Suggest the role of the enzyme DNA methyltransferase (DNMT) in the process of DNA methylation.

(2)

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- *(ii) Some cancers can be caused by the methylation of a tumour suppressor gene called *RASSF1*.

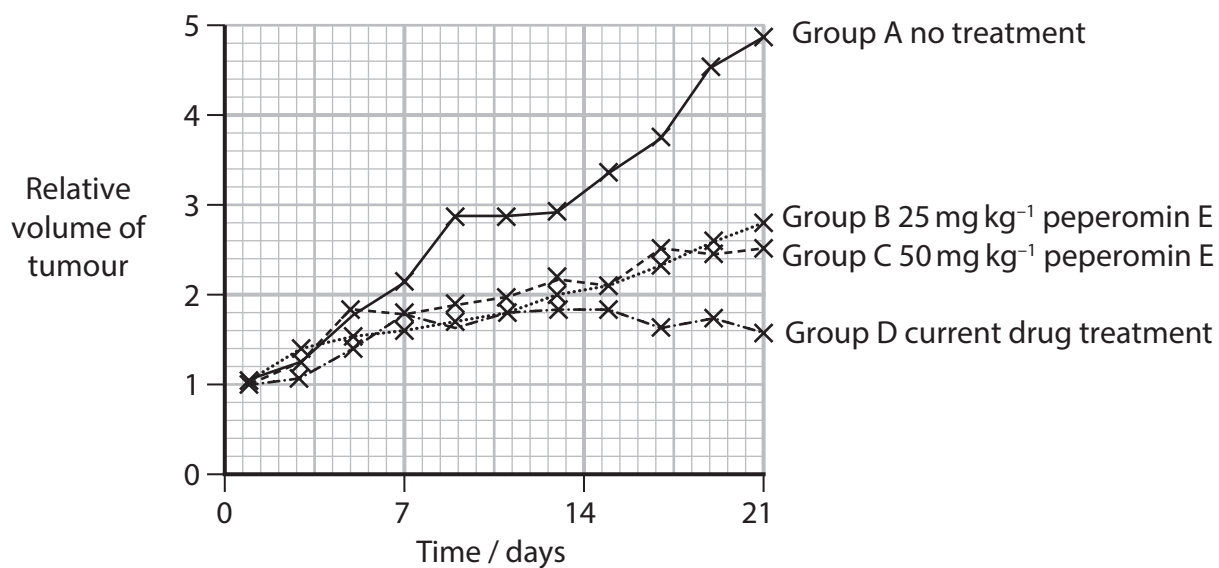
Tumour suppressor genes can slow down cell division or cause cell death.

Peperomin E is a drug that can cause the demethylation of the gene *RASSF1*.

In an investigation different treatments were given to groups of 10 mice.

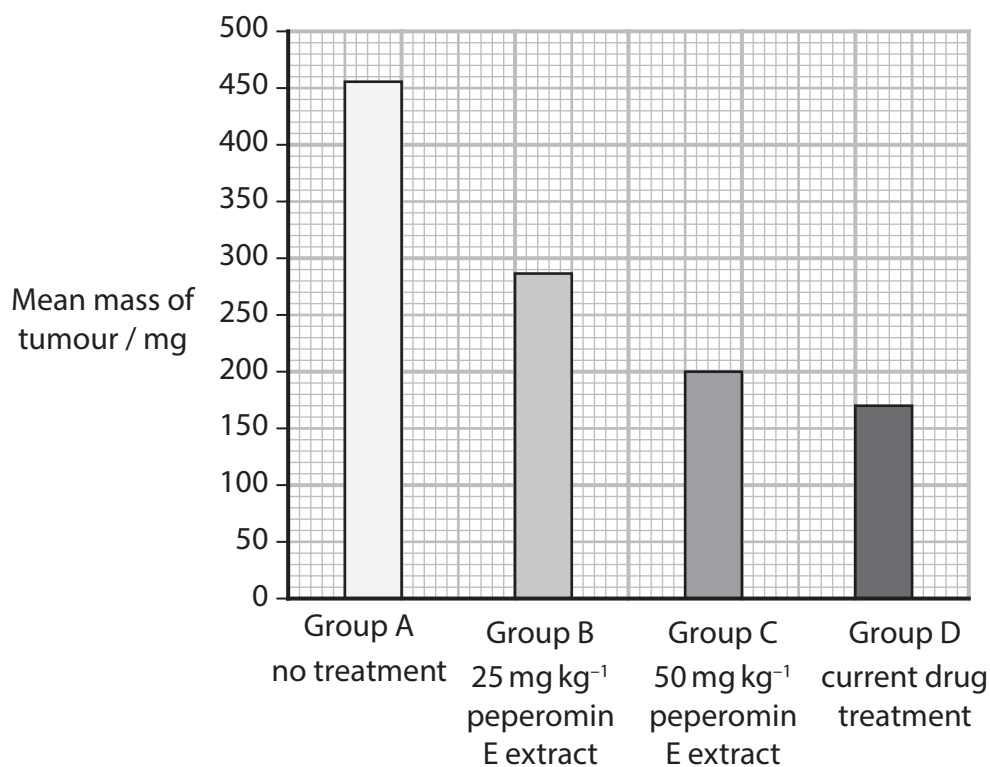
These mice had identical lung cancer tumours at the start of the investigation.

Graph A shows the changes in tumour volume for each group, over 21 days.



Graph A

Graph B shows the mean mass of tumours at 21 days.



Graph B



Explain how treatment with these two drugs affects the activity of the *RASSF1* gene and the changes in tumours.

Use **all** of the information in part (c) and your own knowledge to support your answer.

(6)

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(Total for Question 7 = 14 marks)

